Spend Map — every paid API touchpoint

Exhaustive breakdown of where the app can spend money, what it costs, and every guardrail — audited from the adapter layer, the pipeline, the render farm, and the cost ledger. Mock-first: each provider sits behind an **isXLive()** gate; with no key it returns a $0 mock and the ledger skips it.

7
paid providers (Anthropic, fal.ai, ElevenLabs, Gemini, TwelveLabs, OpenAI, + YouTube quota)

1
ledger — every charge flows through `recordCost` → `cost_ledger`

4
budget caps (per-video · monthly · AI-video $100 · operator 30-day $60)

~$0.15–12
content cost per long-form (Base → Platinum tier)

Providers at a glance — who charges, for what, how much

PROVIDER	LIVE GATE	USED FOR	UNIT PRICE (FROM CODE)
Anthropic Claude **the dominant spender**	`ANTHROPIC_API_KEY`	Scripts, QC judging, editorial guard, fact-check, art direction, seed/beat vision, best-of-N, competitive judge, frame critics, thumbnail pick, data-viz, intel, scout, optimizer, calendar, highlights, choreography, librarian, revision briefs	Opus $15/$75 · Sonnet $3/$15 · Haiku $1/$5 per 1M in/out (`pricing.ts`)
fal.ai	`FAL_KEY`	FLUX still images; AI video clips (Seedance/Kling/Veo/LTX/Wan); Kokoro volume-voice	FLUX $0.003–0.025/img · video $0.022–0.40/sec · Kokoro $0.02/1k ch
ElevenLabs	`ELEVENLABS_API_KEY`	Premium voiceover (`eleven_turbo_v2_5`)	~$0.167/1k chars (env-tunable)
Google Gemini	`GEMINI_API_KEY`	Temporal video understanding (Self-Watch Tier-2); competitor-video perception (intel)	~$0.50/1M tok · `gemini-2.5-flash`
TwelveLabs	`TWELVELABS_API_KEY`	Alt temporal engine (Self-Watch Tier-2), `pegasus1.2`	~$0.05/video
OpenAI	`OPENAI_API_KEY`	Embeddings (`text-embedding-3-small`) — semantic idea-dedup + Studio Memory recall	~$0.02/1M tok · **NOT metered in the ledger**
YouTube Data/Analytics	OAuth	Idea research, publish upload, stats/analytics ingestion	Free — quota-limited, not billed
Pexels	`PEXELS_API_KEY`	Licensed stock footage/photos	$0 (ledgered at zero) — quota, not billing
Telegram	bot token	Approval / alert notifications	Free

Spend by pipeline stage — what fires, and its model tier

STAGE	PAID CALLS THAT FIRE HERE	MODEL / PROVIDER	ROUGH COST
Idea / Intelligence	Idea scout + scoring; idea-gate score + 1 improve round; topic planner; scout chat (YouTube-search loop); Gemini competitor perception (deep scans)	Haiku score · Sonnet plan/scout · Gemini	cents
Script	Outline pre-gate (Haiku); script draft (Sonnet, +extend retry); fact-check (Sonnet + web-search ×≤4); editorial guard (Haiku); QC review; data-viz figure proposal	Sonnet + Haiku + web search	~$0.05–0.30
Assets	Art-director shot plan; prompt refine; FLUX stills (+pixel-check re-roll); seed-still vision gate; beat-relevance vision; variety re-roll; VO synth (ElevenLabs or Kokoro); AI video clips; stick choreography; highlights; pronunciation	Sonnet/Haiku · FLUX · fal-video · EL/Kokoro	tier-driven ($0–$12)
Render (farm)	Vision frame critics (stick + footage); thumbnail vision-pick; data-viz chart verify — all gated behind free ffmpeg media-QC first	Sonnet (VISION_MODEL)	cents

Final / Self-Watch	QC final review (Haiku→Opus escalation); competitive-fit judge (Haiku); Tier-2 temporal pass (Gemini/TwelveLabs) — only if boundaries ≥ escalate-at; best-of-N titles+thumbs	Haiku · Opus · Gemini/TL	cents–$0.05
Auto-fix loop	Re-critique (independent judge); re-choreography / visual re-roll / revision brief; each fix triggers a re-render (the expensive step)	banded Haiku→Sonnet · fal	capped $1/video
Publish / Tracking	YouTube upload + stats/analytics (free, quota)	YouTube API	$0
Nightly / weekly	Optimizer insights (Haiku); librarian lesson synthesis (Haiku, only on promotions); outcome-audit joins; embeddings (OpenAI, unmetered)	Haiku · OpenAI	~$1–5/mo

Per-video content cost by tier (auto-tiers.ts estimates)

TIER	SHORT	LONG-FORM	WHAT YOU GET
Base	~$0	~$0.15–0.40	Script + VO + FLUX stills; no AI video
Economy	~$0.30	~$0.60–1.50	+ ≤3 Seedance-Fast accent clips
Premium	~$1.20	~$2–5	Hero bookends + b-roll ≈1/min (Seedance-2)
Platinum	~$2.50	~$5–12+	Kling heroes + Seedance-2 b-roll

All figures are the constants/estimates in the code (`pricing.ts` , `video-models.ts` , `voice.ts` , `auto-tiers.ts`), not live-billed amounts. Actual cost tracks tokens/seconds used.

Which quality systems cost money

🤖 Anthropic-graded quality

- QC review at every gate (Haiku screen → Opus on borderline)
- Editorial guard · fact-check (+web search) · idea gate
- Seed-still & beat-relevance vision · art-director refine
- Frame critics · thumbnail pick · data-viz verify (render farm)
- Competitive-fit judge · best-of-N title/thumb selection

🎞️ Free / near-free quality

- ffmpeg media-QC (black/frozen/silence/LUFS) — $0, gates the paid vision critics
- Self-Watch timing dimension — structural, no LLM
- Pattern-interrupt & length-band lints — deterministic
- pHash variety check — local compute
- vo_cache / flux_cache reuse — repeats billed $0

The most expensive quality step is re-rendering. Auto-fix and Self-Watch can trigger re-renders (each re-runs generation). That's why re-renders are hard-capped: `maxRenders` + a `$1/video` auto-fix spend cap, and Self-Watch's own re-render cap. The Tier-2 temporal video pass (Gemini/TwelveLabs) is the priciest single quality call and only fires when suspect shot-boundaries ≥ `watchTemporalEscalateAt` .

Every spend guardrail (your protection)

GUARDRAIL	WHAT IT DOES	WHERE
The one ledger	`recordCost` logs every charge to `cost_ledger` (skips ≤$0) and bumps the video's running total for mid-stage checks; `reconcileLedger` repairs drift nightly.	`ledger.ts`
Per-video + monthly caps	`checkBudget` reserves the about-to-spend amount against `budget.perVideoUsd` / `.monthlyUsd` ; a paid stage past a cap holds with the reason.	project budget · `budgetPause`
AI-video ceiling	`VIDEO_MONTHLY_CAP_USD = $100` portfolio-wide; a clip is blocked if spent + estimate exceeds it.	`video-models.ts`
Operator cycle cap	30-day cycle budget (default **$60**); production pauses when reached; one exhaustion alert per cycle; per-slot reservation.	`operator.ts`
Auto-fix cap	`maxRenders` + `spendCapUsd $1` /video; budget-left re-checked before each re-roll.	`autofix.ts`
Fail-closed	If a paid generator (fal) is live but the QC/grading model is not, paid generation is **blocked** — never spend on output you can't grade.	`quality-gates.ts`
Budget reservation	Budget is checked *before* a batch of VO + stills fires, and per-beat with headroom — not just once per stage.	`engine.ts`
Provider circuit-breaker	Terminal errors ("balance exhausted", "payment required", "suspended", quota+billing) trip a flag that stops retries + pauses dependent work + notifies, auto-clearing on a healthy check. **This is what stopped the runaway retry loop.**	`provider-health.ts`
Batch discount	`ANTHROPIC_USE_BATCH=1` routes latency-tolerant fan-outs through the Batches API for a flat **50% off**, stacks with prompt caching.	`anthropic-batch.ts`
Free-reuse caches	`vo_cache` (identical narration re-voices free) · `flux_cache` (identical still prompts re-generate free).	`engine.ts`

Standing infrastructure cost (not per-video)

⚙️ GitHub Actions

- render every 30m (60m cap, heaviest)
- clips · operator · auto-fix · video-intel every 30m
- build-runner every 15m
- intel daily · stats daily · optimizer weekly

☁️ Platform

- Vercel hosting (free → ~$20/mo)
- Supabase Postgres/Storage/Auth (free → ~$25/mo)
- R2 / storage for renders
- Flat monthly, independent of video volume.

🧠 Learning loop

- Embeddings ~$0.02–0.05/mo (unmetered)
- pgvector = $0 incremental
- Nightly librarian ~$1–5/mo
- Total learning-loop spend ≈ $2–7/mo, dominated by the librarian.

- The "idle-cron tax" — runs even when the queue is empty. The circuit-breaker exists to stop these hammering a broken paid provider.

Known accounting gaps (flagged in the audit)

- **OpenAI embeddings are unmetered** — `embedText` computes no cost, so semantic-dedup / memory embedding calls never appear in the ledger (tiny in absolute terms, but the ledger slightly understates true spend).
- **Inline pricing duplication** — ~7 vision/data-viz adapters re-declare the Anthropic price map instead of importing `pricing.ts`, so a price change there won't propagate to those cost estimates.
- **Gemini / TwelveLabs $/unit are hand-set estimates** (`$0.50/1M`, `$0.05/video`), not live-billed figures.
- **Pexels is ledgered at $0** — correct on the free tier; would need a real number on a paid plan.

Bottom line: the dominant variable cost is Anthropic (grading + generation) and fal.ai (video), both hard-capped four ways. The most effective spend levers you control: keep tiers at Base/Economy for unproven formats, raise `watchTemporalEscalateAt` to rarely trigger the temporal pass, flip `ANTHROPIC_USE_BATCH=1` for 50% off overnight fan-outs, and let the caches do their work.